1)

--- Original DataFrame ---

	Employee_ID	Name	Department	Age	Salary	Performance_Rating	Remote
0	101	Alice	HR	25	50000	3.5	True
1	102	Bob	IT	34	75000	4.2	False
2	103	Charlie	IT	29	68000	4.8	True
3	104	David	Finance	42	85000	3.9	True
4	105	Eva	Finance	38	92000	4.5	False
5	106	Frank	IT	50	110000	4.9	False
6	107	Grace	HR	23	48000	3.1	True
7	108	Hannah	Marketing	31	62000	4.0	False

--- 1. Employees with Salary > $70,000 ---

	Employee_ID	Name	Department	Age	Salary	Performance_Rating	Remote
1	102	Bob	IT	34	75000	4.2	False
3	104	David	Finance	42	85000	3.9	True
4	105	Eva	Finance	38	92000	4.5	False
5	106	Frank	IT	50	110000	4.9	False

--- 2. IT Department Top Performers (Rating >= 4.5) ---

	Employee_ID	Name	Department	Age	Salary	Performance_Rating	Remote
2	103	Charlie	IT	29	68000	4.8	True
5	106	Frank	IT	50	110000	4.9	False

--- 3. Employees in HR or Working Remotely ---

	Employee_ID	Name	Department	Age	Salary	Performance_Rating	Remote
0	101	Alice	HR	25	50000	3.5	True
2	103	Charlie	IT	29	68000	4.8	True
3	104	David	Finance	42	85000	3.9	True
6	107	Grace	HR	23	48000	3.1	True

--- 4. Employees in Finance or Marketing ---

	Employee_ID	Name	Department	Age	Salary	Performance_Rating	Remote
3	104	David	Finance	42	85000	3.9	True
4	105	Eva	Finance	38	92000	4.5	False
7	108	Hannah	Marketing	31	62000	4.0	False

--- 5. Names containing 'a' (case insensitive) ---

	Employee_ID	Name	Department	Age	Salary	Performance_Rating	Remote
0	101	Alice	HR	25	50000	3.5	True
2	103	Charlie	IT	29	68000	4.8	True
3	104	David	Finance	42	85000	3.9	True
4	105	Eva	Finance	38	92000	4.5	False
5	106	Frank	IT	50	110000	4.9	False
6	107	Grace	HR	23	48000	3.1	True
7	108	Hannah	Marketing	31	62000	4.0	False

--- 6. Query Method: Age < 35 and Remote == True ---

	Employee_ID	Name	Department	Age	Salary	Performance_Rating	Remote
0	101	Alice	HR	25	50000	3.5	True
2	103	Charlie	IT	29	68000	4.8	True
6	107	Grace	HR	23	48000	3.1	True

=== ORIGINAL DATAFRAME ===

	Employee_ID	Name	Department	Age	Salary	Performance_Rating	Remote	Join_Date
0	101	Alice Smith	HR	25.0	50000	3.5	True	2021-01-15
1	102	Bob Jones	IT	34.0	75000	4.2	False	2019-03-22
2	103	Charlie Brown	IT	29.0	68000	4.8	True	2020-07-11
3	104	David Villa	Finance	42.0	85000	3.9	True	2018-11-05
4	105	Eva Green	Finance	38.0	92000	4.5	False	2017-06-30
5	106	Frank Wright	IT	50.0	110000	4.9	False	2015-01-10
6	107	Grace Lee	HR	23.0	48000	3.1	True	2022-08-19
7	108	Hannah Abbott	Marketing	31.0	62000	4.0	False	2020-02-14
8	109	Ian Malcolm	IT	NaN	95000	4.6	True	2019-10-01
9	110	Julia Roberts	Marketing	45.0	105000	3.8	False	2016-04-12

--- 1. AND Logic: Department == 'IT' & Salary > 70k ---

	Employee_ID	Name	Department	Age	Salary	Performance_Rating	Remote	Join_Date
1	102	Bob Jones	IT	34.0	75000	4.2	False	2019-03-22
5	106	Frank Wright	IT	50.0	110000	4.9	False	2015-01-10
8	109	Ian Malcolm	IT	NaN	95000	4.6	True	2019-10-01

--- 2. OR Logic: Department == 'HR' | Remote == True ---

	Employee_ID	Name	Department	Age	Salary	Performance_Rating	Remote	Join_Date
0	101	Alice Smith	HR	25.0	50000	3.5	True	2021-01-15
2	103	Charlie Brown	IT	29.0	68000	4.8	True	2020-07-11
3	104	David Villa	Finance	42.0	85000	3.9	True	2018-11-05
6	107	Grace Lee	HR	23.0	48000	3.1	True	2022-08-19
8	109	Ian Malcolm	IT	NaN	95000	4.6	True	2019-10-01

--- 3. NOT Logic: NOT HR & Performance >= 4.0 ---

	Employee_ID	Name	Department	Age	Salary	Performance_Rating	Remote	Join_Date
1	102	Bob Jones	IT	34.0	75000	4.2	False	2019-03-22
2	103	Charlie Brown	IT	29.0	68000	4.8	True	2020-07-11
4	105	Eva Green	Finance	38.0	92000	4.5	False	2017-06-30
5	106	Frank Wright	IT	50.0	110000	4.9	False	2015-01-10
7	108	Hannah Abbott	Marketing	31.0	62000	4.0	False	2020-02-14
8	109	Ian Malcolm	IT	NaN	95000	4.6	True	2019-10-01

--- 4. Mixed Logic: (IT or Finance) & Remote ---

	Employee_ID	Name	Department	Age	Salary	Performance_Rating	Remote	Join_Date
2	103	Charlie Brown	IT	29.0	68000	4.8	True	2020-07-11
3	104	David Villa	Finance	42.0	85000	3.9	True	2018-11-05
8	109	Ian Malcolm	IT	NaN	95000	4.6	True	2019-10-01

--- 5. Helpers: Age between 25-40 & Dept in ['IT', 'Marketing'] ---

	Employee_ID	Name	Department	Age	Salary	Performance_Rating	Remote	Join_Date
1	102	Bob Jones	IT	34.0	75000	4.2	False	2019-03-22
2	103	Charlie Brown	IT	29.0	68000	4.8	True	2020-07-11
7	108	Hannah Abbott	Marketing	31.0	62000	4.0	False	2020-02-14

--- 6. String + Numeric: Name contains 'an' & Salary >= 60k ---

	Employee_ID	Name	Department	Age	Salary	Performance_Rating	Remote	Join_Date
5	106	Frank Wright	IT	50.0	110000	4.9	False	2015-01-10

=== ORIGINAL DATAFRAME ===

	Name	Department	Salary	Age	Performance_Rating
0	Alice	IT	75000.0	34	4.2
1	Bob	IT	75000.0	29	4.8
2	Charlie	Finance	85000.0	42	3.9
3	David	IT	68000.0	29	4.0
4	Eva	Finance	92000.0	38	4.5
5	Frank	HR	50000.0	25	3.5
6	Grace	HR	48000.0	23	3.1
7	Hannah	Finance	NaN	31	4.0

--- 1. Sorted by Department (A-Z) & Salary (High to Low) ---

	Name	Department	Salary	Age	Performance_Rating
4	Eva	Finance	92000.0	38	4.5
2	Charlie	Finance	85000.0	42	3.9
7	Hannah	Finance	NaN	31	4.0
5	Frank	HR	50000.0	25	3.5
6	Grace	HR	48000.0	23	3.1
0	Alice	IT	75000.0	34	4.2
1	Bob	IT	75000.0	29	4.8
3	David	IT	68000.0	29	4.0

--- 2. Sorted by Age (Youngest First) & Performance (Highest First) ---

	Name	Department	Salary	Age	Performance_Rating
6	Grace	HR	48000.0	23	3.1
5	Frank	HR	50000.0	25	3.5
1	Bob	IT	75000.0	29	4.8
3	David	IT	68000.0	29	4.0
7	Hannah	Finance	NaN	31	4.0
0	Alice	IT	75000.0	34	4.2
4	Eva	Finance	92000.0	38	4.5
2	Charlie	Finance	85000.0	42	3.9

--- 3. Sorted by Salary (Ascending) with Missing Values FIRST ---

	Name	Department	Salary	Age	Performance_Rating
7	Hannah	Finance	NaN	31	4.0
6	Grace	HR	48000.0	23	3.1
5	Frank	HR	50000.0	25	3.5
3	David	IT	68000.0	29	4.0
0	Alice	IT	75000.0	34	4.2
1	Bob	IT	75000.0	29	4.8
2	Charlie	Finance	85000.0	42	3.9
4	Eva	Finance	92000.0	38	4.5

--- 4. Sorted with Clean/Reset Index Numbers ---

	Name	Department	Salary	Age	Performance_Rating
0	Eva	Finance	92000.0	38	4.5
1	Charlie	Finance	85000.0	42	3.9
2	Hannah	Finance	NaN	31	4.0
3	Frank	HR	50000.0	25	3.5

=== ORIGINAL DATAFRAME ===

	Employee_ID	Name	Department	Age	Salary	Bonus
0	101	Alice	IT	25.0	50000.0	NaN
1	102	Bob	HR	NaN	75000.0	NaN
2	103	Charlie	NaN	29.0	68000.0	NaN
3	104	David	Finance	42.0	NaN	NaN
4	105	None	IT	38.0	92000.0	NaN
5	106	Frank	Finance	NaN	110000.0	NaN
6	107	Grace	HR	23.0	48000.0	NaN

==
1. DETECTING MISSING VALUES
==

--- Boolean Mask (True = Missing) ---

	Employee_ID	Name	Department	Age	Salary	Bonus
0	False	False	False	False	False	True
1	False	False	False	True	False	True
2	False	False	True	False	False	True
3	False	False	False	False	True	True
4	False	True	False	False	False	True

--- Rows where 'Salary' is missing ---

	Employee_ID	Name	Department	Age	Salary	Bonus
3	104	David	Finance	42.0	NaN	NaN

--- Rows where 'Salary' is NOT missing ---

	Employee_ID	Name	Department	Age	Salary	Bonus
0	101	Alice	IT	25.0	50000.0	NaN
1	102	Bob	HR	NaN	75000.0	NaN
2	103	Charlie	NaN	29.0	68000.0	NaN
4	105	None	IT	38.0	92000.0	NaN
5	106	Frank	Finance	NaN	110000.0	NaN
6	107	Grace	HR	23.0	48000.0	NaN

==
2. COUNTING MISSING VALUES
==

```
--- Missing values count PER COLUMN ---
Employee_ID    0
Name           1
Department     1
Age            2
Salary         1
Bonus          7
dtype: int64

--- Percentage (%) of missing values PER COLUMN ---
Employee_ID      0.00
Name            14.29
Department      14.29
Age             28.57
Salary          14.29
Bonus          100.00
dtype: float64

--- Total missing cells in DataFrame: 12 ---
--- Rows containing at least one missing value: 7 out of 7 ---
```

==
3. REMOVING MISSING VALUES
==

--- A. Drop rows with ANY missing value ---

	Employee_ID	Name	Department	Age	Salary	Bonus

--- B. Drop rows where 'Salary' OR 'Department' is missing ---

	Employee_ID	Name	Department	Age	Salary	Bonus
0	101	Alice	IT	25.0	50000.0	NaN
1	102	Bob	HR	NaN	75000.0	NaN
4	105	None	IT	38.0	92000.0	NaN
5	106	Frank	Finance	NaN	110000.0	NaN
6	107	Grace	HR	23.0	48000.0	NaN

--- C. Drop columns that are 100% missing (e.g., 'Bonus') ---

	Employee_ID	Name	Department	Age	Salary
0	101	Alice	IT	25.0	50000.0
1	102	Bob	HR	NaN	75000.0
2	103	Charlie	NaN	29.0	68000.0
3	104	David	Finance	42.0	NaN
4	105	None	IT	38.0	92000.0
5	106	Frank	Finance	NaN	110000.0
6	107	Grace	HR	23.0	48000.0

--- D. Drop rows with fewer than 5 valid values ---

	Employee_ID	Name	Department	Age	Salary	Bonus
0	101	Alice	IT	25.0	50000.0	NaN
6	107	Grace	HR	23.0	48000.0	NaN

=== ORIGINAL DATAFRAME ===

	Employee_ID	Name	Department	Salary	City
0	101	Alice	IT	75000	NY
1	102	Bob	HR	60000	LA
2	102	Bob	HR	60000	LA
3	103	Charlie	Finance	85000	Chicago
4	104	David	IT	75000	NY
5	105	Eva	Finance	90000	Houston
6	105	Eva	Finance	95000	Houston

==
1. DETECTING & COUNTING DUPLICATES
==

--- A. Boolean Mask for Exact Duplicate Rows ---

```
0    False
1    False
2     True
3    False
4    False
5    False
6    False
dtype: bool
```

--- Total Exact Duplicate Rows: 1 ---

--- View All Occurrences of Duplicated Rows ---

	Employee_ID	Name	Department	Salary	City
1	102	Bob	HR	60000	LA
2	102	Bob	HR	60000	LA

--- Duplicate entries based on 'Employee_ID' only: 2 ---

==
2. REMOVING DUPLICATE DATA
==

--- A. Remove Exact Duplicates (Keep First) ---

	Employee_ID	Name	Department	Salary	City
0	101	Alice	IT	75000	NY
1	102	Bob	HR	60000	LA
3	103	Charlie	Finance	85000	Chicago
4	104	David	IT	75000	NY
5	105	Eva	Finance	90000	Houston
6	105	Eva	Finance	95000	Houston

--- B. Deduplicate by 'Employee_ID' (Keep FIRST seen) ---

	Employee_ID	Name	Department	Salary	City
0	101	Alice	IT	75000	NY
1	102	Bob	HR	60000	LA
3	103	Charlie	Finance	85000	Chicago
4	104	David	IT	75000	NY
5	105	Eva	Finance	90000	Houston

--- C. Deduplicate by 'Employee_ID' (Keep LAST seen) ---

	Employee_ID	Name	Department	Salary	City
0	101	Alice	IT	75000	NY
2	102	Bob	HR	60000	LA
3	103	Charlie	Finance	85000	Chicago

=== ORIGINAL DATAFRAME ===

	Employee_ID	Name	Department	Location	Salary	Bonus
0	101	Alice	IT	NY	75000	5000
1	102	Bob	IT	LA	68000	4000
2	103	Charlie	Finance	NY	85000	8000
3	104	David	Finance	Chicago	92000	9000
4	105	Eva	IT	NY	110000	12000
5	106	Frank	HR	LA	50000	3000
6	107	Grace	HR	NY	48000	2500
7	108	Hannah	Marketing	Chicago	62000	4500
8	109	Ian	IT	LA	95000	10000
9	110	Julia	Marketing	NY	105000	11000

==
1. OVERALL SUMMARY STATISTICS
==

--- Summary statistics (.describe()) ---

	Employee_ID	Salary	Bonus
count	10.00	10.00	10.00
mean	105.50	79000.00	6900.00
std	3.03	21924.11	3502.38
min	101.00	48000.00	2500.00
25%	103.25	63500.00	4125.00
50%	105.50	80000.00	6500.00
75%	107.75	94250.00	9750.00
max	110.00	110000.00	12000.00

```
Total Salary Mass (Sum):  $790,000
Average Salary (Mean):    $79,000.00
Median Salary:            $80,000.00
Max Salary:               $110,000
Min Salary:               $48,000
```

==
2. GROUPBY AGGREGATION (BY DEPARTMENT)
==

--- Average Salary per Department ---

	Department	Salary
0	Finance	88500.0
1	HR	49000.0
2	IT	87000.0
3	Marketing	83500.0

--- Employee Count per Department ---

	Department	Employee_Count
0	Finance	2
1	HR	2
2	IT	4
3	Marketing	2

==
3. ADVANCED AGGREGATIONS WITH .agg()
==

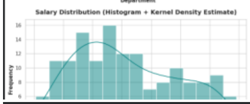
--- Department Summary (Custom Named Aggregations) ---

	Department	Employee_Count	Avg_Salary	Max_Salary	Total_Bonus

=== SAMPLE DATASET HEAD ===

	Department	Years_Experience	Performance_Score	Remote	Date_Joined	Salary
0	Finance	1	5.0	On-Site	2022-01-02	58248
1	Marketing	1	3.7	On-Site	2022-01-09	53980
2	HR	3	3.2	Remote	2022-01-16	53316
3	HR	14	4.7	Remote	2022-01-23	108739
4	IT	2	4.4	On-Site	2022-01-30	68965

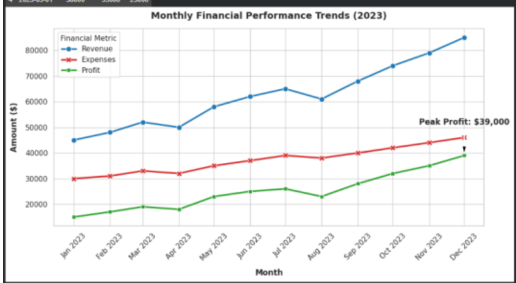
```
/tmp/ipykernel_3106/1334090629.py:36: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  ax = sns.barplot(data=df, x='Department', y='Salary', estimator=np.mean, errorbar=None, palette='viridis')
```



=== SAMPLE TIME-SERIES DATA ===

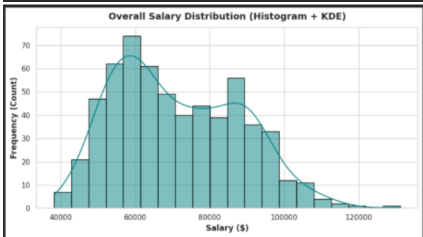
	Date	Revenue	Expenses	Profit
0	2023-01-01	45000	30000	15000
1	2023-02-01	48000	31000	17000
2	2023-03-01	52000	33000	19000
3	2023-04-01	50000	32000	18000
4	2023-05-01	58000	35000	23000

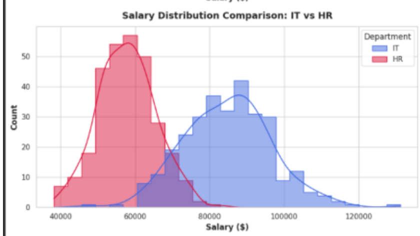
colab.research.google.com – To exit full screen, press Esc

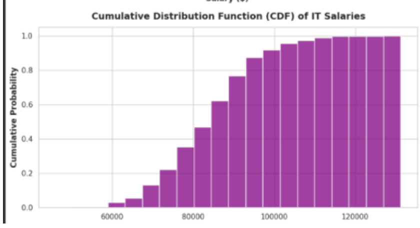


=== AVERAGE SALARY BY DEPARTMENT ===

	Department	Salary
2	IT	90760.583333
0	Finance	82735.375000
4	Sales	74786.666667
3	Marketing	67305.500000
1	HR	61161.666667



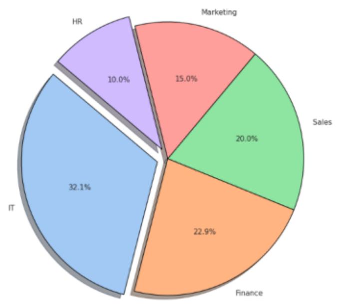


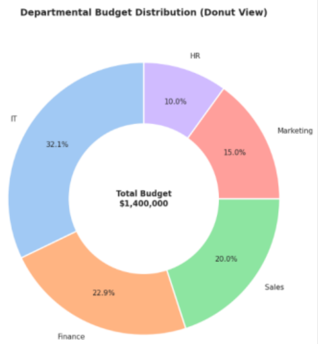



Salary vs. Experience by Dept, Work Mode & Performance
Salary ($)
Years of Experience
Department
Marketing
IT
Finance
Performance_Rating
2.5
3.0
3.5
4.0
4.5
5.0
Work_Mode
On-Site
Remote

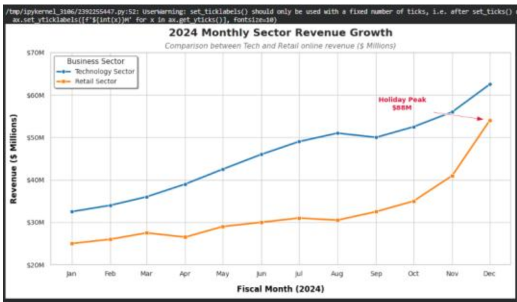

Experience vs. Salary with Linear Regression Trendline
Salary ($)
Years of Experience


Salary vs. Experience Breakdown Across Departments
Department = Marketing
Department = IT
Department = Finance
Salary ($)
Work_Mode
On-Site
Remote


HR
10.0%
Marketing
15.0%
Sales
20.0%
Finance
22.9%
IT
32.1%


Departmental Budget Distribution (Donut View)
HR
10.0%
Marketing
15.0%
Sales
20.0%
Finance
22.9%
IT
32.1%
Total Budget
$1,400,000

```
/tmp/ipykernel_3106/2392255447.py:52: UserWarning: set_ticklabels() should only be used with a fixed number of ticks, i.e. after set_ticks() or using a FixedLocator.
  ax.set_yticklabels([f'${int(x)}M' for x in ax.get_yticks()], fontsize=10)
```




Employee Performance & Compensation Dashboard
1. Monthly Avg Salary Trend
Avg Salary ($)
Date
2. Avg Salary by Department
Salary ($)
Finance
HR
IT
Marketing
Department
3. Performance Score Distribution
Employee Count
Performance Score
4. Experience vs Salary Correlation
Dept
Marketing
IT
Finance
HR
Salary ($)
Years of Experience
Full-Width Top Panel: Executive Revenue Summary
Salary ($)
Date
Departmental Salary Ranges
Salary
Marketing
IT
Finance
HR
Department
Salary Density Curves
Density
Department
Marketing
IT
Finance
HR
Salary

=== ORIGINAL DATAFRAME ===

	Employee_ID	Name	Department	Salary	Join_Date
0	101	Alice Smith	IT	75000	2021-01-15
1	102	Bob Jones	Finance	85000	2019-03-22
2	103	Charlie Brown	IT	68000	2020-07-11
3	104	David Villa	HR	52000	2022-08-19
4	105	Eva Green	Finance	92000	2017-06-30

Successfully wrote DataFrame to 'employees_data.csv'!

=== IMPORTED DATAFRAME ===

	Employee_ID	Name	Department	Salary	Join_Date
0	101	Alice Smith	IT	75000	2021-01-15
1	102	Bob Jones	Finance	85000	2019-03-22
2	103	Charlie Brown	IT	68000	2020-07-11
3	104	David Villa	HR	52000	2022-08-19
4	105	Eva Green	Finance	92000	2017-06-30

```
--- Column Data Types ---
Employee_ID     int64
Name           object
Department     object
Salary          int64
Join_Date      object
dtype: object
```

=== ADVANCED IMPORTED DATAFRAME ===

Employee_ID	Name	Salary	Join_Date
101	Alice Smith	75000	2021-01-15
102	Bob Jones	85000	2019-03-22
103	Charlie Brown	68000	2020-07-11
104	David Villa	52000	2022-08-19
105	Eva Green	92000	2017-06-30

```
--- Advanced Data Types (Join_Date is datetime64) ---
Name                 object
Salary                int64
Join_Date    datetime64[ns]
dtype: object
```